Sheet 63 Thu 29 Apr. 2027 *Proof by contradiction and by cases*

1. Warm-up · Quick [DIV] What is the parity of $n(n+1)$ for an integer n?

   Ans. Always even: one of the two consecutive integers is even.

2. Core · Proof by cases [DEM] Show that $n(n+1)$ is even for every integer n.

   Ans. If n is even, the product contains an even factor. If n is odd, $n+1$ is even. In both cases, the product is even.

3. Core + · Contradiction [DEM] Show by contradiction that the sum of a rational number r and an irrational number a is irrational.

   Ans. Suppose $r+a=q\in\mathbb{Q}$; then $a=q-r\in\mathbb{Q}$ (difference of two rationals): contradiction with a being irrational.

4. Review · Formula [IMG] $g(x)=2x+5$. Image of -1? Preimage of 11?

   Ans. $g(-1)=3$; $2x+5=11\Leftrightarrow x=3$.

5. Thread · Formula → graph [REG] Plot the points with abscissae $-2\,;\,-1\,;\,0\,;\,1\,;\,2$ on the curve of $f(x)=x^2$; give their ordinates.

   Ans. $4\,;\,1\,;\,0\,;\,1\,;\,4$: the points are symmetric about the vertical axis.

Sheet 64 Mon 3 May 2027 *Logical connectives, contrapositive and counter-example in context*

1. Warm-up · Quick [DEM] Write the negation of "$x>2$" and then of "$x>2$ or $x<-1$".

   Ans. $x\leqslant 2$; $x\leqslant 2$ and $x\geqslant -1$, that is $-1\leqslant x\leqslant 2$.

2. Core · Method [DEM] Write the contrapositive of "if n^2 is even, then n is even".

   Ans. "If n is odd, then n^2 is odd" (equivalent to the original statement).

3. Core + · Counter-example [DEM] Disprove: "every prime number is odd".

   Ans. Counter-example: 2 is prime and even.

4. Review · True/false [PNP] True or false? Justify. The perimeter of a square is proportional to the length of its side.

   Ans. True. $P=4c$: constant of proportionality 4.

5. Thread · Reading code [ALG] What does $h(3)$ return? Write $h(x)$ as an expression.

   Ans. $h(3)=5\times 5=25$; $h(x)=(x+2)^2$.

Probability: conditioning and weighted trees

Sheet 65 Tue 4 May 2027 *Probability model and equally likely outcomes*

1. Warm-up · Quick [PRB] A fair coin is tossed three times. How many outcomes are there?

   Ans. $2^3=8$ equally likely outcomes.

2. Core · Method [PRB] Biased die: $P(6)=0.25$ and the other five faces are equally likely. Calculate $P(1)$.

   Ans. $\dfrac{1-0.25}{5}=0.15$.

3. Core + · Critical thinking [PRB] "It will rain or it will not rain tomorrow: so there is a one in two chance." Is this equally-likely model justified?

   Ans. No: having two outcomes does not make them equally likely. Equal likelihood is an assumption of the model that must be justified.

4. Review · Quick [PER] Which shapes make up the net of a square-based pyramid?

   Ans. 1 square and 4 (isosceles) triangles.

5. Thread · Translate [LIT] Each side of a square of side x is increased by 3. Express the increase in area in terms of x.

   Ans. $(x+3)^2-x^2=6x+9$.

Sheet 66 Mon 10 May 2027 *Law of large numbers (informal version): simulation*

1. Warm-up · Quick [STA] In 100 tosses, 47 are "heads". What is the relative frequency of "heads"?

   Ans. 0.47.

2. Core · Interpret [PRB] Why, in 10 tosses of a fair coin, can the frequency of "heads" stray from 0.5? What happens with 10 000 tosses?

   Ans. Sampling fluctuation: a small sample has high variability. For large samples, the frequency stabilises near the probability 0.5 (law of large numbers).

3. Core + · True/false [PRB] True or false? Justify. Tossing a fair coin 1 000 times gives exactly 500 "heads".

   Ans. False. The number obtained is close to 500, rarely exactly 500.

4. Review · Problem [MOD] Price including tax $P(x) = 1.2\,x$ (20 % VAT). Calculate $P(250)$ and the preimage of 480.

   Ans. $P(250) = 300$; $1.2x = 480 \Leftrightarrow x = 400$.

5. Thread · Method [EQ] Solve in $\mathbb{R}$: $\dfrac{2x-1}{3} = \dfrac{x+4}{2}$.

   Ans. $x = 14$

   Cross-multiplication: $2(2x-1) = 3(x+4)$.

Sheet 67 Tue 11 May 2027 *Python simulation of a repeated random experiment*

1. Warm-up · Quick [PRB] $P(A) = 0.35$. Calculate $P(\overline{A})$.

   Ans. $1 - 0.35 = 0.65$.

2. Core · Reading code [ALG] What does this program compute? Roughly what value will it display?

```
from random import randint
compte = 0
for i in range(1000):
    if randint(1, 6) == 6:
        compte = compte + 1
print(compte / 1000)
```

   Ans. The frequency of a 6 in 1 000 rolls of a die; about $\dfrac{1}{6} \approx 0.167$.

3. Core + · Method [PRB] Fair 12-sided die with faces numbered 1 to 12. Probability of getting a prime number?

   Ans. Primes: 2 ; 3 ; 5 ; 7 ; 11: $\dfrac{5}{12}$ (1 is not prime).

4. Review · Construction [CON] M lies on the perpendicular bisector of $[AB]$. What can be said about MA and MB? How is it constructed with compasses?

   Ans. $MA = MB$. Construction: two arcs of the same radius (greater than $\frac{1}{2}AB$) centred at A and B; their intersections define the perpendicular bisector.

5. Thread · Recognise [SOL] What is the shape of the cross-section of a cube by a plane parallel to one of its faces?

   Ans. A square with the same side as the face.

Sheet 68 Thu 13 May 2027 *Conditional probability $P_A(B)$*

1. Warm-up · Quick [PRB] Among 25 canteen users, 15 are girls. A canteen user is chosen at random: probability that this is a girl?

   Ans. $\dfrac{15}{25} = 0.6$.

2. Core · Method [PRB] In a class of 30 pupils, 12 play football, of whom 5 also play handball. A pupil is chosen at random: calculate $P_F(H)$, where F: "plays football", H: "plays handball".

   Ans. $P_F(H) = \dfrac{\text{Card}(F \cap H)}{\text{Card}(F)} = \dfrac{5}{12}$.

3. Core + · Method [PRB] In an equally likely model, $\text{Card}(A) = 50$ and $\text{Card}(A \cap B) = 20$. Calculate

$P_A(B)$.

Ans. $\dfrac{20}{50} = 0.4$.

4. Review · Quick [NUM] Put in increasing order: $\dfrac{7}{4}$, $-\dfrac{3}{2}$, 0.5.

   Ans. $-\dfrac{3}{2} < 0.5 < \dfrac{7}{4}$ $(-1.5 < 0.5 < 1.75)$.

5. Thread · True/false [DEM] True or false? Justify. A quadrilateral with two opposite sides parallel is a parallelogram.

   Ans. False. Counter-example: a trapezium (not a parallelogram) has only one pair of parallel sides.

Sheet 69 Tue 18 May 2027 *Conditional probability and two-way tables*

1. Warm-up · Quick [PRB] In the following table, what is the total of the "sport" row?

	Girls	Boys	Total
Sport	18	22	…
No sport	12	8	20
Total	30	30	60

   Ans. $18 + 22 = 40$.

2. Core · Method [PRB] A pupil is chosen at random from this table. Calculate $P(S)$, $P_F(S)$ and $P_S(F)$ (F: girl, S: does sport).

   Ans. $P(S) = \dfrac{40}{60} = \dfrac{2}{3}$; $P_F(S) = \dfrac{18}{30} = 0.6$; $P_S(F) = \dfrac{18}{40} = 0.45$.

3. Core + · True/false [PRB] True or false? Justify. In this table, $P_F(S) = P_S(F)$.

   Ans. False. $0.6 \neq 0.45$: the reference sets (30 girls and 40 sports participants) differ.

4. Review · Quick [PUI] Calculate $5^0 + 2^{-1}$.

   Ans. $1 + \dfrac{1}{2} = \dfrac{3}{2}$.

5. Thread · Method [REP] $ABCD$ is a parallelogram with $A(1;1)$, $B(5;2)$, $C(6;5)$. Find D.

   Ans. The diagonals have the same midpoint: $D = A + C - B = (2;4)$. Check: $\overrightarrow{AB} = \overrightarrow{DC} = (4;1)$.

Sheet 70 Thu 20 May 2027 *Weighted trees: construction*

1. Warm-up · Quick [PRB] What is the sum of the probabilities of the branches leaving a single node?

   Ans. 1.

2. Core · Method [PRB] In a population, 30 % have a disease M. A test is positive (T) for 90 % of those with the disease and for 5 % of those without it. Complete the branch probabilities of the tree.

   Ans. $P(M) = 0.3$, $P(\overline{M}) = 0.7$; $P_M(T) = 0.9$, $P_M(\overline{T}) = 0.1$; $P_{\overline{M}}(T) = 0.05$, $P_{\overline{M}}(\overline{T}) = 0.95$.

3. Core + · Translate [PRB] Translate into symbols: "among those with the disease, 90 % test positive".

   Ans. $P_M(T) = 0.9$: branch $M \to T$; do not write $P(T) = 0.9$.

4. Review · Quick [CAL] Calculate $\dfrac{2}{3} \div \dfrac{4}{9}$.

   Ans. $\dfrac{2}{3} \times \dfrac{9}{4} = \dfrac{18}{12} = \dfrac{3}{2}$.

5. Thread · Problem [DIV] Three lighthouses flash every 6, 8 and 9 seconds. They light up together at $t = 0$. When is the next simultaneous flash?

   Ans. PPCM$(6\,;\,8\,;\,9) = 72$: after 72 s.

Sheet 71 Mon 24 May 2027 *Weighted trees: multiplying along a path*

1. Warm-up · Quick [CAL] Calculate 0.3×0.9.

   Ans. 0.27.

2. Core · Method [PRB] Using the tree from sheet 70, calculate $P(M \cap T)$ and $P(\overline{M} \cap T)$.

   Ans. $P(M \cap T) = 0.3 \times 0.9 = 0.27$; $P(\overline{M} \cap T) = 0.7 \times 0.05 = 0.035$.
   We multiply along a path: $P(M \cap T) = P(M) \times P_M(T)$.

3. Core + · Method [PRB] Deduce $P(T)$.

   Ans. Two paths lead to T: $0.27 + 0.035 = 0.305$.

4. Review · Factorise [DEV] Factorise $x^2 - 9$.

   Ans. $(x-3)(x+3)$.

5. Thread · Quick [GEO] The side $[BC]$ of length 13 cm is the hypotenuse of a right-angled triangle ABC. What is the radius of its circumscribed circle?

   Ans. The midpoint of the hypotenuse is the centre: radius $= 13 \div 2 = 6.5$ cm.

Sheet 72 Tue 25 May 2027 *From table to tree and symbolic notation*

1. Warm-up · Quick [PRB] Express $\dfrac{30}{60}$ as a probability.

   Ans. 0.5.

2. Core · Method [PRB] From the table on sheet 69, build the tree (F or G, then S or $\overline{S}$): give $P(F)$, $P_F(S)$ and $P_G(S)$.

   Ans. $P(F) = 0.5$; $P_F(S) = \dfrac{18}{30} = 0.6$; $P_G(S) = \dfrac{22}{30} = \dfrac{11}{15}$.

3. Core + · Translate [PRB] Write symbolically and calculate "the probability of being a boy who does not do sport".

   Ans. $P(G \cap \overline{S}) = \dfrac{8}{60} = \dfrac{2}{15}$; or $0.5 \times \dfrac{8}{30}$.

4. Review · Quick [QP] A car travels at constant speed: 150 km in 2 h. What distance in 5 h?

   Ans. $75 \times 5 = 375$ km.

5. Thread · Statement → formula [REG] A taxi charges a pick-up fee of 2 000 Ar then 1 500 Ar per km. Express the price $p(x)$ for x km, then calculate $p(8)$.

   Ans. $p(x) = 1\,500x + 2\,000$; $p(8) = 14\,000$ Ar.

Sheet 73 Thu 27 May 2027 *Reversing conditioning: $P_A(B)$ and $P_B(A)$*

1. Warm-up · Quick [PRB] What is the difference between $P_M(T)$ and $P_T(M)$?

   Ans. $P_M(T)$: probability of testing positive given that one has the disease; $P_T(M)$: of having the disease given that one tests positive.

2. Core · Method [PRB] Using the data from sheet 70 ($P(M \cap T) = 0.27$ and $P(T) = 0.305$), calculate $P_T(M)$.

   Ans. $P_T(M) = \dfrac{P(M \cap T)}{P(T)} = \dfrac{0.27}{0.305} \approx 0.885$.

3. Core + · Method [PRB] Rare disease (1 % of the population); the test is positive for 99 % of those with the disease and 5 % of those without. Calculate $P_T(M)$.

   Ans. $P(M \cap T) = 0.0099$; $P(\overline{M} \cap T) = 0.99 \times 0.05 = 0.0495$; $P(T) = 0.0594$; $P_T(M) = \dfrac{0.0099}{0.0594} = \dfrac{1}{6} \approx 0.17$.
   Counter-intuitive result: a test that is "99 % reliable" is not enough.

4. Review · Graph [IMG] Curve of f: